Exercise 12.1

1. The following distribution represents the scores achieved by a group of chemistry students in the chemistry laboratory.

Scores	24 – 28	29 – 33	34 – 38	39 – 43	44 – 48	49 – 53	Total
No. of students	3	6	12	23	15	6	65

Answer the following questions.

(i) What is the upper limit of the last class?

$$Upper\ limit\ of\ the\ last\ class = 53$$

(ii) What is the lower limit of the class 39 – 43?

$$Lower\ limit\ of\ the\ class\ 39 - 43 = 39$$

(iii) What is the midpoint of the class (34 – 38)?

$$Midpoint = \frac{Lower\ Limit + Upper\ Limit}{2}$$

$$Midpoint = \frac{34 + 38}{2}$$

$$Midpoint = \frac{72}{2}$$

$$Midpoint = 36$$

(iv) What are the class frequencies of the classes 29 – 33 and 44 – 48?

$$Class\ frequency\ of\ 29 - 33 = 6$$

$$Class\ frequency\ of\ 44 - 48 = 15$$

(v) What is the size of the class limits in the above frequency distribution?

$$Class\ size = Upper\ limit - Lower\ limit + 1$$

$$Class\ size = 28 - 24 + 1$$

$$Class\ size = 5$$

(vi) In which class or group does minimum number of students fall?

$$Minimum\ frequency = 3\ (in\ class\ 24 - 28)$$

(vii) What is the lower limit of the class having 15 as its class frequency?

Class with frequency 15 is $44 - 48$. So,

$$Lower\ limit = 44$$

(viii) What is the number of students having scores between 24 and 43?

Classes included: 24 – 28, 29 – 33, 34 – 38, 39 – 43

$$Number\ of\ students = 3 + 6 + 12 + 23$$

$$Number\ of\ students = 44$$

2. For a school staff, the following expenditures (rupees in hundred) are required for the repair of chairs.
145, 152, 153, 156, 158, 160, 146, 152, 155, 159, 161, 163, 165, 147, 148, 151, 154, 156, 158, 160, 144, 167, 151, 150, 152, 149, 145, 153, 152, 155
Prepare a frequency distribution by tally bar method by taking number of classes 8, also write down what are the frequencies of the last three classes?

: Let

$$Expenditures\ for\ the\ repair\ of\ chairs = X$$

$$Minimum\ value = X_{min} = 144$$

$$Maximum\ value = X_{max} = 167$$

$$Range = X_{max} - X_{min}$$

$$Range = 167 - 144$$
$$Range = 23$$
$$Number\ of\ classes = k = 8$$
$$Size\ of\ class\ interval = h = \frac{Range}{k}$$
$$h = \frac{23}{8}$$
$$h = 2.875$$
$$h \approx 3$$

Class Interval	Tally Marks	f
$144 - 146$	||||	4
$147 - 149$	|||	3
$150 - 152$	|||||||	7
$153 - 155$	|||||	5
$156 - 158$	||||	4
$159 - 161$	||||	4
$162 - 164$	|	1
$165 - 167$	||	2